

Network Poisoning Attacks

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Attacks and Exploits

Objective: 4.2: Given a scenario, perform network attacks using the appropriate tools.

Video Overview

This video introduces you to network poisoning attacks, specifically focusing on Address Resolution Protocol (ARP) and Domain Name System (DNS) poisoning. We'll explore how these techniques exploit vulnerabilities in older network protocols to manipulate network traffic. Understanding these attacks is important for anyone pursuing a career in cybersecurity or network administration, as it provides insight into how malicious actors can intercept or redirect data, leading to significant security breaches in real-world scenarios. By learning about these methods, you'll be better equipped to identify and mitigate such threats, protecting networks and sensitive information.

Key Topics Covered

- **Network manipulation and poisoning:** ARP and DNS record poisoning to intercept or modify traffic.
- **Older protocols with weak security:** Built for convenience, not robust protection.
- **ARP Poisoning (Spoofing):** Redirects traffic by faking MAC-IP mappings and intercepts traffic for legitimate devices.
- **Goals of ARP Poisoning:** Enables on-path attacks (intercept, modify, drop traffic) and can cause session hijacking or Denial of Service (DoS).
- **ARP Poisoning Tools:** Ettercap, Bettercap, Impacket, Responder.
- **Domain Name System (DNS) Poisoning:** Exploits the domain name system to redirect victims to malicious sites and provides misinformation.
- **Goals of DNS Attacks:** Redirect traffic to fake login portals (credential harvesting) and modify host files or DNS resolver cache.
- **DNS Poisoning Tools:** Ettercap, Bettercap, Responder (via LLMNR and NBT-NS).

Student Notes Section